

GUYMON, OK, USA

WMO: 722095

Lat: 36.682N

Lon: 101.505W

Elev: 952

StdP: 90.40

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 03030

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-12.9	-10.3	-18.9	0.8	-7.1	-16.6	1.0	-2.9	15.2	6.2	13.8	4.1	4.0	10	0.620

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	38.6	19.9	37.2	19.7	35.7	19.6	22.4	31.9	21.8	31.6	21.3	31.2	6.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	16.5	24.7	19.2	15.7	24.3	18.7	15.2	24.1	70.7	31.9	68.4	31.6	66.3	31.1	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.5	11.8	10.7	DB	-17.7	41.2	2.9	1.6	-19.7	42.3	-21.4	43.3	-23.0	44.1	-25.1	45.3	(4)
(5)			WB	-18.3	23.7	2.7	0.6	-20.2	24.1	-21.8	24.4	-23.4	24.8	-25.4	25.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	2.7	3.9	9.3	13.6	18.9	25.0	27.0	25.7	22.1	14.7	8.2	2.5		
	DBStd	10.04	5.59	6.12	5.45	5.08	4.80	3.34	2.87	3.00	3.91	5.10	5.32	5.50		
	HDD10.0	864	233	183	77	23	2	0	0	0	0	17	94	237		
	HDD18.3	2323	486	404	283	156	52	1	0	0	13	133	305	491		
	CDD10.0	2514	5	13	55	129	278	450	527	488	363	162	39	4		
	CDD18.3	931	0	1	2	13	70	201	269	230	126	20	1	0		
	CDH23.3	12179	2	24	122	354	1066	2561	3459	2668	1488	383	52	1		
	CDH26.7	6499	0	6	33	134	499	1419	2046	1477	733	144	8	0		
(14) Wind	WSAvg	4.9	4.5	4.8	5.3	5.9	5.2	5.3	4.6	4.2	4.8	4.8	4.7	4.5		
(15) Precipitation	PrecAvg	465	13	13	27	35	55	73	75	77	31	44	12	15		
	PrecMax	554	43	37	115	105	122	174	164	141	80	153	90	98		
	PrecMin	347	1	5	0	2	9	8	22	10	2	0	0	0		
	PrecStd	52	12	9	26	25	30	42	34	39	23	45	19	22		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.7	26.5	30.1	33.4	36.9	40.8	40.7	39.4	37.4	33.7	27.9	21.9		
		MCWB	9.6	10.2	12.7	13.4	15.9	18.0	20.2	20.1	18.8	15.8	12.1	9.2		
	2%	DB	19.6	22.5	27.1	30.3	34.1	37.9	38.6	37.9	35.3	30.6	24.6	17.6		
		MCWB	7.9	9.1	11.8	13.0	16.5	18.8	20.3	20.3	18.2	15.9	11.1	7.8		
	5%	DB	15.8	19.3	23.5	27.8	32.0	35.9	37.3	36.1	33.3	27.8	21.4	15.0		
		MCWB	6.3	7.8	10.7	13.2	16.3	19.3	20.4	20.3	18.7	15.0	10.2	6.5		
	10%	DB	12.6	15.3	20.3	24.5	29.4	34.0	35.7	34.0	31.2	24.8	18.0	12.1		
		MCWB	5.0	6.1	9.7	12.6	16.2	19.5	20.4	20.2	18.5	13.8	8.9	5.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	11.7	15.9	17.5	20.7	22.9	23.3	23.1	21.8	20.1	13.9	10.8		
		MCDB	21.1	23.0	24.2	24.2	28.4	32.9	33.0	31.1	30.7	26.0	22.2	16.6		
	2%	WB	8.5	9.7	13.8	15.8	19.6	21.9	22.3	22.5	20.8	18.1	12.3	8.6		
		MCDB	17.6	21.1	20.8	23.9	27.6	31.8	32.4	31.2	30.0	24.2	20.7	17.3		
	5%	WB	6.8	8.2	12.0	14.7	18.6	21.2	21.8	21.8	20.1	16.7	11.0	6.9		
		MCDB	15.3	18.6	20.7	22.6	26.7	30.9	32.0	31.2	29.2	24.4	20.6	14.0		
	10%	WB	5.2	6.6	10.4	13.3	17.5	20.6	21.3	21.2	19.4	15.2	9.5	5.4		
		MCDB	12.2	15.2	19.8	22.2	25.8	30.0	31.8	30.6	28.1	22.8	17.6	11.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.3	15.7	16.6	16.0	15.7	15.5	15.1	14.4	15.2	15.7	16.0	14.2		
		MCDBR	20.8	22.6	21.8	21.5	19.7	18.3	17.6	17.4	18.4	20.2	20.9	19.0		
	5% WB	MCWBR	11.6	11.7	10.1	8.9	7.3	5.9	4.7	4.5	5.9	8.2	10.4	11.0		
		MCWBR	19.8	21.6	19.0	16.6	16.2	15.2	15.1	14.6	14.7	15.8	18.3	17.9		
(40) Clear-Sky Solar Irradiance	taub		0.257	0.265	0.293	0.322	0.341	0.373	0.378	0.364	0.337	0.300	0.271	0.256		
		taud	2.577	2.567	2.492	2.428	2.436	2.385	2.412	2.420	2.479	2.551	2.561	2.582		
	Ebn at Noon		960	988	979	960	940	905	897	906	918	931	931	938		
		Edh at Noon	73	84	100	113	114	120	116	113	101	85	74	69		
(44) All-Sky Solar Radiation	RadAvg		2.86	3.69	4.95	6.16	7.04	7.54	7.36	6.47	5.51	4.12	3.13	2.52		
	RadStd		0.26	0.36	0.42	0.34	0.42	0.37	0.32	0.33	0.26	0.45	0.28	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.47	N/A	N/A	N/A	N/A		
(47) Regional (18 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+75	N/A		

Nomenclature: See separate page